NCERT Summary: Topic 20: Land, Soil & Water Resources of India — Raw Notes

Source Integration

- NCERT Class 8 — Resources and Development, Chapter 2: Land, Soil, Water, Natural Vegetation and Wildlife Resources
- NCERT Class 10 — Contemporary India II, Chapter 1: Resources and Development & Chapter 3: Water Resources

Core Factual & Conceptual Architecture

1. Land Resources & Land Use Dynamics

- Scarcity & Distribution: Land covers 30 percent of the globe, but rugged relief, thick forests, and extreme climates leave 70 percent sparsely populated. Plains and river valleys host the highest population concentrations.
- Land-Use Purposes: Allocated across agriculture, forestry, mining, housing, and industry, governed by physical factors (topography, soil, climate) and human factors (population, technology).
- Ownership Categories: Differentiated between private land (individual ownership) and community land (common property resources for collective village use).
- Degradation & Hazards: Rising demographic pressure drives common land encroachment, triggering land degradation, desertification, and catastrophic landslides. Mitigation involves hazard mapping, drainage control, and afforestation.

2. Soil Resources, Formation & Typology

- Genesis & Profile: Formed via weathering of parent rocks over centuries, creating distinct strata: topsoil (humus), subsoil, weathered rock, and parent bedrock.
- Indian Soil Typology:
  - Alluvial Soil: Most widespread and fertile; found in northern plains and deltas (split into older Bhangar and newer Khadar).
  - Black Soil (Regur): Lava-derived Deccan plateau soil, rich in clay, retaining moisture and ideal for cotton.
  - Red and Yellow Soils: Developed on crystalline igneous rocks under lower rainfall.
  - Laterite Soil: Formed via intense leaching in high-temperature, high-rainfall tracts, suited for tea and cashew after treatment.
  - Arid and Desertic Soils: Sandy, saline tracts found in western Rajasthan.
- Conservation Practices: Agronomic and mechanical controls including mulching, contour barriers, rock dams, terrace farming, intercropping, contour ploughing, and shelter belts.

3. Water Resources & Management

- Freshwater Availability: Three-fourths of Earth is water, but oceanic salinity renders most of it unusable. Only 2.7 percent is freshwater, and of that, 70 percent is frozen in ice caps, leaving roughly 1 percent accessible.
- Scarcity Drivers: Driven by natural precipitation fluctuations and human over-exploitation, industrial pollution, and domestic waste discharge.
- Management Solutions: Rainwater harvesting, drip and sprinkler irrigation, effluent treatment, and revival of traditional systems like guls, kuls, khadins, and bamboo drip irrigation. Multi-purpose river projects balance irrigation and power generation, though accompanied by ecological displacement concerns.

✍ Notes & Updates